What is a quantum computer?

QUANTUM COMPUTING | Fundamentals | As of September 24, 2026

A next-generation computer that uses quantum-mechanical properties in computation itself. Working differently from conventional machines, it is expected to solve certain problems efficiently.

Conventional versus quantum

A conventional bit is either “0” or “1”. A quantum bit uses superposition of “0” and “1”.

Three properties behind computation

Superposition: represents multiple states in a superposition.
Entanglement: states of multiple qubits are strongly related.
Interference: probability amplitudes reinforce or cancel each other, making the desired answer more likely.

What is expected?

Drug discovery, new materials, quantum-chemistry simulation, logistics and production-planning optimization, and financial computation. Research targets some problems whose computational workload grows rapidly on conventional machines. This is not a universal PC that speeds up every calculation.

Japan's listed quantum-computing companies

Hardware and physical-machine R&D | As of September 24, 2026

Fujitsu / 6702 | Superconducting hardware and software

Jointly developed a 256-qubit superconducting quantum computer with RIKEN. Plans to install and make a 1,000-qubit machine available in fiscal 2026. Targets more than 10,000 physical qubits around fiscal 2030.

NTT / 9432 | Photonic quantum computing

Joint research with OptQC. Targets a 10,000-qubit-class machine by fiscal 2027 and a fault-tolerant million-qubit-class machine by fiscal 2030. Applies optical communications and IOWN technologies to quantum computation.

Hitachi / 6501 | Silicon qubits

Participates in JST's Moonshot R&D program. Works toward 100-qubit devices by fiscal 2028 and 1,000-qubit devices by fiscal 2030. Compatibility with existing semiconductor technology is a feature.

NEC / 6701 | Superconducting and quantum annealing

Develops quantum annealing machines using superconducting parametron devices. Also researches superconducting-circuit integration, connected to future fault-tolerant general-purpose quantum computers.

Toshiba also researches superconducting quantum computers, but is excluded from this listed-company roster because it is unlisted.

Four points for reading the quantum theme

Separate technology from business | As of September 24, 2026

1 Check the computing approach

Superconducting, photonic, silicon and neutral-atom approaches require different equipment and materials. Compare companies' R&D with these differences in mind.

2 Separate the computer from supporting technology

The computer, controls, cooling, optical components and software have different roles. Also distinguish quantum cryptography, which serves a different purpose from quantum computation.

3 Do not compare qubit counts alone

Error rates, fidelity, error correction and connectivity between qubits matter too. Examine the accuracy and scale at which useful computation can run.

4 Assess the distance to practical use and sales

Research results and sales contributions are separate. Check progress against development targets and the quantum business's share of total company performance.

Japanese companies use different approaches

Fujitsu: superconducting; NTT: photonic; Hitachi: silicon; NEC: superconducting and quantum annealing. No dominant approach is established; several may coexist for different uses.